

Leveraging Edge AI for AI Decarbonization & Accessibility

A Dual System for Consumer-Facing Impact Tracking and Edge-Based Inference to Reduce the Energy, Water, and Carbon Footprint of AI

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Abstract

AI has a real environmental cost now, and the numbers are no longer small. In 2025, global data centres pulled an estimated 448 terawatt-hours of electricity, with AI workloads accounting for roughly a fifth of that draw and a carbon footprint of about 189 million tonnes of CO₂-equivalent^[1]. Water tells a similar story: AI-related data-centre withdrawals reached close to 264 billion gallons globally that same year^[2]. Both figures are still climbing.

This report covers two things I designed and built in response. The first is a Chrome extension for real-time inference tracking, a working prototype, already deployed, that estimates the energy, water, and carbon cost of a person's AI interactions as they happen and turns that into something relatable, with the aim of nudging more responsible day-to-day AI use. The second is an edge computing platform, presented here as a design recommendation rather than a finished build. Its purpose is to optimise AI models for on-device inference, so processing happens closer to the user instead of inside a centralised, energy-hungry cloud data centre.

To test whether the edge idea actually holds up, I modelled a smart-agriculture sensor network under both cloud and edge processing. Moving the same workload from cloud to edge cut projected annual energy use from 7,013.26 kWh to 438.01 kWh, and carbon emissions from 3,506.63 kg CO₂e to 219.01 kg CO₂e, a reduction of roughly 93.75%. That figure lines up closely with independently published research showing edge inference can cut energy and water use for generative AI workloads by over 90% relative to cloud processing^[12]. Together, the two systems point at the same underlying argument: making AI sustainable and accessible means doing two things at once, showing people the footprint they generate, and moving the heaviest computation away from centralised infrastructure and out to the edge, particularly for the energy- and connectivity-constrained regions that cloud-dependent AI currently serves worst.

Keywords: *AI energy consumption, AI water footprint, edge computing, decarbonization, digital carbon tracking, AI accessibility, sustainable AI, edge inference*

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Chapter 1: Project Proposal

1.1 Introduction

Artificial intelligence, at its core, is an attempt to replicate human intelligence in a machine. Before the term itself existed, coined later by John McCarthy, computers could only follow instructions given to them directly ^[3]. AI has moved through three broad eras since then: simple rule-based systems, transformer architectures, and now the large language models built on transformers, with GPT-3 among the first at real scale. Each shift brought a jump in the energy needed to build and run these systems, across development, deployment, and everyday use.

That energy use splits into two stages worth separating. Training, the *developmental phase*, is shaped by model size, the number of parameters, the compute available, how long training runs, the infrastructure it runs on, and how the data itself gets processed ^[4]. *Inference* is the other stage: once a model is trained and deployed, energy is consumed every time it is queried, scaling with usage rather than with development choices. GPT-3, at roughly 175 billion parameters, used about 1,287 megawatt-hours to train ^[1], comparable to around 120 years of an average American household's electricity use, or about 4.29 billion Google searches ^[4]. At the level of a single prompt, the comparison usually cited is a full smartphone charge, or leaving a refrigerator running for about half an hour ^[5].

This project started from that same point back in 2024, but the picture has sharpened considerably since. By 2025, data centres worldwide were consuming an estimated 448 terawatt-hours of electricity a year, with AI responsible for roughly a fifth of it and 189 million tonnes of CO₂-equivalent emissions attached to that. If data centres were a country, they would be the eleventh-largest electricity consumer in the world ^[1]. Less discussed, but just as real, is water: AI data centres used an estimated 264 billion gallons globally in 2025 for cooling alone ^[2], and researchers at the University of California, Riverside estimate that every 20 to 50 conversational AI prompts costs roughly 500 millilitres of water once on-site cooling and the water intensity of the electricity behind it are both counted ^[6]. The UN University Institute for Water, Environment and Health projects AI-related water withdrawals could reach 9.3 trillion litres a year by 2030 if current growth holds ^[9]. I have treated energy, carbon, and water as one connected footprint throughout this report rather than as separate numbers, which is a change from how earlier drafts of this project approached it.

1.2 Problem Statement

AI keeps working its way into daily life and enterprise workflows, and the computational resources behind it have grown just as fast, along with the electricity and water that comes with them, much of it still drawn from grids that lean heavily on fossil fuels. That footprint is not spread evenly. It concentrates in the regions that host hyperscale data centres, while communities in energy- and connectivity-constrained areas, the ones who could benefit most from AI-enabled tools, remain the least able to reach them, because most current AI infrastructure simply assumes reliable grid power and broadband connectivity.

Two gaps follow from that. The first sits on the consumption side: people generating this footprint every time they query an AI system have almost no way to see its cost. A wave of browser-based carbon and energy trackers has emerged since 2025^[22]–^[29], which is genuinely useful, but most of them stop at observation. They report a footprint after the fact and leave it there, without connecting it to anything the user, developer, or industry could actually act on. The second gap sits on the production side. AI holds real promise for climate and resource-management work, but less energy-intensive techniques such as edge computing remain under-adopted, and where edge-AI tooling does exist, it is rarely designed with low-connectivity, low-power deployment in mind, even though that is precisely where it would matter most.

1.3 Hypothesis

My working hypothesis is that the energy, water, and carbon footprint of large-scale AI systems can be meaningfully reduced without giving up usefulness, through two things working together: real-time, consumer-facing visibility into what an individual AI interaction actually costs, which should encourage more deliberate use, and a shift of inference workloads off centralised cloud infrastructure and onto optimised edge devices, which cuts the energy required per inference and removes the dependence on high-bandwidth connectivity, in turn improving accessibility in energy- and connectivity-constrained regions.

1.4 Main Research Objective

This project looks at the relationship between AI's computational footprint and its environmental and accessibility consequences, and builds two things in response: a working browser-based tool that tracks and communicates the footprint of AI use as it happens, and a design for an edge-based inference platform that reduces that footprint at its source while extending AI's reach into underserved regions.

1.5 Specific Objectives

- Quantify AI's energy, water, and carbon footprint across development and inference, using current published data.
- Build and deploy a Chrome extension that estimates and displays the footprint of a user's AI interactions in real time, using relatable comparisons rather than raw numbers.
- Design an edge computing platform that optimises AI models for on-device inference, and test its potential impact through a quantified case study.
- Compare the projected environmental performance of cloud versus edge processing for a representative use case.
- Position both tools against the current landscape of carbon-tracking software and edge-AI research, to see where they actually add something new.
- Recommend next steps, with real attention to accessibility and decarbonization in energy-undersupplied regions, much of Sub-Saharan Africa included.

1.6 Scope

- Energy, water, and carbon consumption analysis across AI development and inference.
- Design and evaluation of a consumer-facing inference-tracking Chrome extension.
- Design and case-study validation of an edge-computing optimisation platform.
- A representative deployment scenario (smart-agriculture sensor monitoring) and a representative edge target (image-based crop-health classification on a Raspberry Pi).
- Geographic framing centred on accessibility for energy-undersupplied and low-connectivity regions.

1.7 Technical Requirements

- Data collection tools for AI energy, water, and carbon coefficients published in peer-reviewed and industry research.
- Analytical tools for modelling cloud versus edge energy and carbon outcomes.
- A browser extension development environment (Chrome Manifest V3) for the tracking tool.
- Model optimisation resources (quantization, pruning, lightweight-model conversion) for the edge platform.

1.8 Significance of Study

This work matters for a few reasons, and I will try to keep this section short rather than pad it out. AI's electricity and water draw is outpacing public awareness of it, and the gap between what AI promises and what it costs in resources is widest exactly where people can least afford that cost. What I have built here is also meant to be proactive rather than reactive: a tool that steps in before or during consumption, not one that only reports on it afterward. And the edge computing case is, I think, as much about democratization as it is about efficiency. The same architectural shift that cuts AI's footprint also removes its dependence on the centralised cloud infrastructure that many energy-undersupplied regions simply do not have.

Chapter 2: Literature Review

Energy sits right at the intersection of what makes AI useful and what it costs the environment, and that cost has grown steadily as adoption has scaled. This chapter works through that footprint, the evidence for edge computing as a way to reduce it, what already exists in consumer-facing tracking tools, and the accessibility angle specific to energy-undersupplied regions, before landing on the gap this project is trying to close.

2.1 Energy Consumption in AI and Its Carbon Footprint

Training energy scales with model size, the compute behind it, and infrastructure choice. The table below pulls together published figures for a few representative model classes.

Model Size (Parameters)	Computational Resources	Training Duration (Hours)	Infrastructure	Training Energy (MWh)	Carbon Footprint
0.34B (BERT-Large)	16GB VRAM	96	Cloud	1.5	≈1,400
7B	8× NVIDIA V100	336	Cloud (efficient)	50	N/A
40B	64× NVIDIA V100	672	Cloud (efficient)	200	N/A
175B (GPT-3)	1,024× NVIDIA V100	816–1,344	Cloud (standard)	1,287	≈1.1M

Table 1: Comparative energy consumption of large-scale AI models by parameter count. Sources: [2], [4].

Electricity demand at the data-centre level has moved well past what any single model's training run suggests. Global data centres used around 448 TWh of electricity in 2025, with AI accounting for roughly a fifth of that, and a carbon footprint of 189 million tonnes CO₂e, enough that offsetting it would take something like 3.2 billion tree seedlings grown for a decade^[1]. If AI's share of data-centre electricity climbs to 40% by 2030, as one scenario has it, AI alone could account for around 3% of global electricity use^[1].

2.2 The Water Footprint of AI

Water has only recently become part of this conversation. Data centres use it two ways: directly, mostly for evaporative cooling, and indirectly, through the water intensity of the electricity that powers them. Cooling alone can account for 30–40% of a facility's total energy use^[10]. Li et al. estimated that training GPT-3 in Microsoft's U.S. data centres used roughly 5.4 million litres of water in total, 700,000 litres of that on-site for cooling, and that global AI-related water withdrawals could

reach 4.2–6.6 billion cubic metres a year by 2027, or four to six times Denmark's annual consumption^[6]. At the level of individual use, UC Riverside researchers put it at roughly 500 millilitres of water for every 20 to 50 conversational prompts, once indirect, generation-side water use is counted^[6]. Zoomed out to the aggregate level, AI data centres used an estimated 264 billion gallons (nearly one trillion litres) of water globally in 2025, about 550 million gallons a day, comparable to the entire global bottled-water industry's daily output^[2]. A UN University report projects that figure could reach 9.3 trillion litres a year by 2030^[9]. None of this is easy to verify independently, either: a 2025 University of Toronto study found that none of six major hyperscale operators reviewed disclosed site-specific water data in a consistent, comparable format^[2], which is itself part of the transparency problem this project is working against.

2.3 Edge Computing as a Decarbonization Strategy

There is a solid and growing body of research behind edge computing (processing data on or near the device generating it, rather than shipping it to a centralised cloud data centre) as a real decarbonization lever, not just a latency fix. One case study comparing cloud and edge processing for generative AI inference found edge platforms could cut energy use by over 90% while reducing carbon emissions and water consumption by more than 80%^[12]. Task-offloading research using real-world data found up to 83.3% reduction in total carbon footprint through optimal scheduling between edge and cloud resources^[13], and a HAL/INRIA study of distributed compute nodes found fully distributed edge architectures use 14–25% less energy than centralised alternatives, mostly by cutting out long-distance data transmission^[16]. A carbon-aware edge inference framework showed a 22.9–32.2% reduction in emissions compared with running everything centrally, with almost no scheduling overhead^[11]. On the network side, computing at the edge within a 5G network has been shown to cut the associated CO₂ footprint by up to 50%, since data transmission itself is a real emissions source, alongside the radio access network and the servers^[14]. Reviews of edge AI in energy-system applications add that it also reduces upstream data traffic, defers network upgrades that would otherwise be needed, and supports greater integration of renewable and distributed energy resources^[15].

2.4 Consumer-Facing Carbon and Energy Awareness Tools: What Already Exists

When I first sketched out the Chrome extension in 2024, there was not much dedicated consumer tooling for AI-specific footprint tracking. That has changed fast. By 2026 there is a whole category of browser extensions doing this: some estimate energy use from message length and show it inline on ChatGPT^[23], some keep a running energy-and-water total visible on the page with everyday

comparisons like phone charges or millilitres of water^[24], some treat the token itself as the unit of environmental cost and pair real-time token telemetry with grid carbon-intensity data^[29], and some lean more toward offsetting than pure measurement^[26]. A related, AI-agnostic category tracks the carbon footprint of general web browsing by category or by domain^{[22] [27]}.

This emerging market backs up the premise this project started from, but a gap runs through almost all of it: these tools are observational. They measure and display a footprint, sometimes suggest offsetting it, but none of them connect that measurement to anything concrete on the production side. None point the user, developer, or platform toward an architectural fix, edge deployment being the obvious one, that would actually reduce the footprint rather than just accounting for it after the fact. Closing that specific gap is what pairing a tracking extension with an edge-computing platform recommendation is meant to do.

2.5 AI Accessibility and Energy Infrastructure in Underserved Regions

The accessibility case for edge computing is strongest exactly where centralised cloud AI is least reliable. As of 2024, 4G coverage reached only around 65% of Africa's population, with much of the remainder on legacy 3G networks or entirely disconnected^[17]. That is a real constraint for any AI application that needs live cloud connectivity, and a direct argument for edge deployment, which does not depend on a constant high-bandwidth link. Where infrastructure is developing, it is developing unevenly. Nairobi's IXAfrica campus went live in 2025 as East Africa's first AI-ready, GPU-dense data-centre infrastructure, built through a partnership with Safaricom as part of a \$500 million regional AI-infrastructure investment^[19]. The global edge-computing market, worth roughly \$168 billion in 2025, is projected to grow fastest in the Middle East and Africa region, at 17–19% annually, even though the region currently holds only about 5% of global edge server capacity^[20]. Power is the real bottleneck: Africa's data-centre power demand is forecast to grow at a 9% compound annual rate and reach 2 GW by 2030, against a global installed base projected at 374 GW^[20]. Off-grid, edge-based monitoring is already filling part of that gap. Edge computing gateways deployed in Ugandan electricity networks, for instance, process voltage, current, and load data locally, specifically because cellular coverage is patchy and constant cloud connectivity cannot be assumed^[18]. Separately, research into federated learning on low-resource edge devices, Raspberry Pi-class hardware included, has shown that meaningful model performance is achievable without the large centralised compute clusters AI development usually assumes, which supports edge deployment as a democratization strategy and not only an efficiency one^[21].

2.6 Summary of the Literature Review, Gaps, and Contribution to Knowledge

A few gaps come out of this review. The fast-growing category of AI carbon-tracking tools is still almost entirely observational, and stops short of connecting a measured footprint to anything actionable on the production side. The technical case for edge computing as a decarbonization strategy is strong, but it lives mostly in simulation- and infrastructure-level research, and rarely makes it into a form that is legible to a consumer or ready to deploy. And research on AI accessibility in energy- and connectivity-constrained regions, Africa especially, is currently concentrated on macro-level infrastructure investment rather than the application-layer tooling that individuals and small organisations could actually use today. What this project contributes sits across those three gaps: a working, consumer-facing tracking tool that makes AI's footprint visible in the moment of use, paired with an edge-computing platform recommendation, validated through a representative case study, that gives that visibility somewhere concrete to lead: toward AI infrastructure that is both lower-footprint and more accessible in the regions current cloud-centric AI serves worst.

Chapter 3: Requirement Gathering, Analysis & Methodology

3.1 Summary of System Requirement Specification

Two related but different systems came out of this project. The first, and the only one actually built and deployed, is a Chrome extension that performs real-time inference tracking. As someone interacts with an AI platform, the extension estimates the energy, water, and carbon cost of that interaction and surfaces it as an immediate, relatable figure, with the explicit goal of shaping more responsible AI use. The second is an edge computing platform, still at the design-recommendation stage: a platform that would let developers optimise AI models for edge-device deployment, cutting reliance on energy-intensive cloud inference and extending AI access into places that cannot count on constant, high-bandwidth cloud connectivity. Its feasibility is shown here through a quantified case study rather than a live deployment.

3.2 Methodology

- Desk research and literature review, establishing current, cited coefficients for AI energy use, grid carbon intensity, and water intensity per unit of compute, drawn from peer-reviewed and industry sources (Chapter 2).
- Extension development: building the Chrome extension to detect interaction with supported AI platforms, apply a footprint-estimation methodology to that interaction, and render the result in real time and cumulatively.
- Case-study modelling, to quantify the relative energy, water, and carbon cost of cloud versus edge processing for a representative deployment scenario, and validate the edge platform's projected impact ahead of a full build.
- Comparative positioning: benchmarking the extension against the current landscape of similar tools identified in the literature review, to see where it actually adds distinct value.
- Synthesis, consolidating findings into a recommended platform architecture and roadmap (Chapters 4 through 6).

3.3 Functional Requirements

System	Functional Requirement
Inference Tracking Extension	Detect when a user is interacting with a supported AI platform.

System	Functional Requirement
Inference Tracking Extension	Estimate energy, water, and carbon cost per interaction from interaction length/complexity.
Inference Tracking Extension	Display real-time and cumulative (daily/weekly/all-time) footprint data.
Inference Tracking Extension	Express estimates as relatable, everyday equivalents (e.g. share of a phone charge).
Edge Computing Platform (proposed)	Accept a trained model and target edge hardware profile as input.
Edge Computing Platform (proposed)	Apply optimisation (quantization, pruning, lightweight conversion) suited to the target device.
Edge Computing Platform (proposed)	Output a deployment-ready package for on-device inference, online or offline.

Table 6: Functional requirements for the inference tracking extension and edge platform.

3.4 Non-Functional Requirements

- Privacy: interaction data is processed locally in the browser wherever possible, and the extension does not transmit personal usage data to third parties.
- Performance: tracking should not introduce perceptible latency into a user's interaction with AI platforms.
- Accessibility: the extension and its estimation methodology need to remain usable on low-bandwidth connections, and the edge platform needs to target low-power, low-memory devices such as the Raspberry Pi class of hardware.
- Transparency: the estimation methodology should be disclosed to users, since footprint figures are estimates rather than direct metering, and trust depends on that distinction being clear.

3.5 System Usability & Performance

The extension is meant to be passive. Nobody has to change how they use AI platforms for tracking to happen, and feedback comes through in plain, non-technical language with real-world comparisons rather than raw units alone, which is the same approach most of the comparable tools in Chapter 2 take. The goal is for footprint feedback to show up within the same interaction cycle as

the AI's own response, so the cost of a query is still legible while that query is fresh in the user's mind.

3.6 Standard Development Tools Used

The extension is built on the Chrome Extension platform using Manifest V3, with browser-local storage for usage history and no server-side collection of individual query content. The edge platform recommendation in Chapter 4 is scoped around widely available, low-power edge hardware (Raspberry Pi-class devices) and lightweight inference runtimes such as TensorFlow Lite and ONNX, consistent with the optimisation techniques reviewed in Chapter 2. This section reflects the general architecture; exact library versions and any additional tooling used in the deployed build can be filled in here.

3.7 Programming Languages & Frameworks

- JavaScript/TypeScript and the Chrome Extension APIs, for the inference tracking extension.
- Python, with TensorFlow Lite, ONNX, or PyTorch mobile tooling, for edge model optimisation.

Chapter 4: System Design

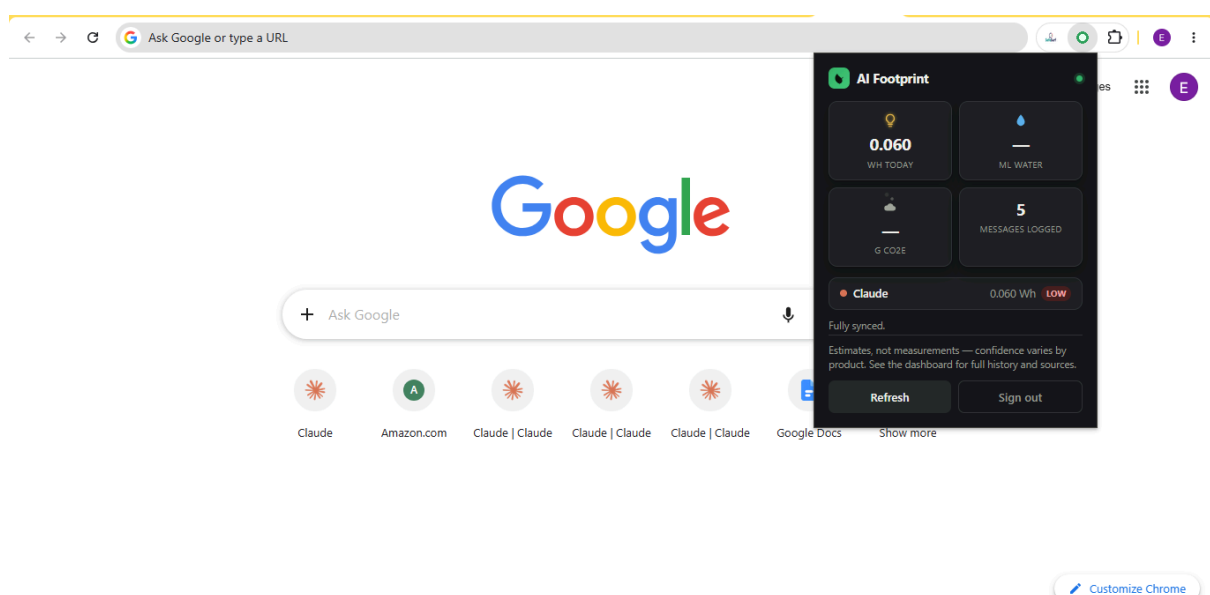
4.1 System Architecture Overview

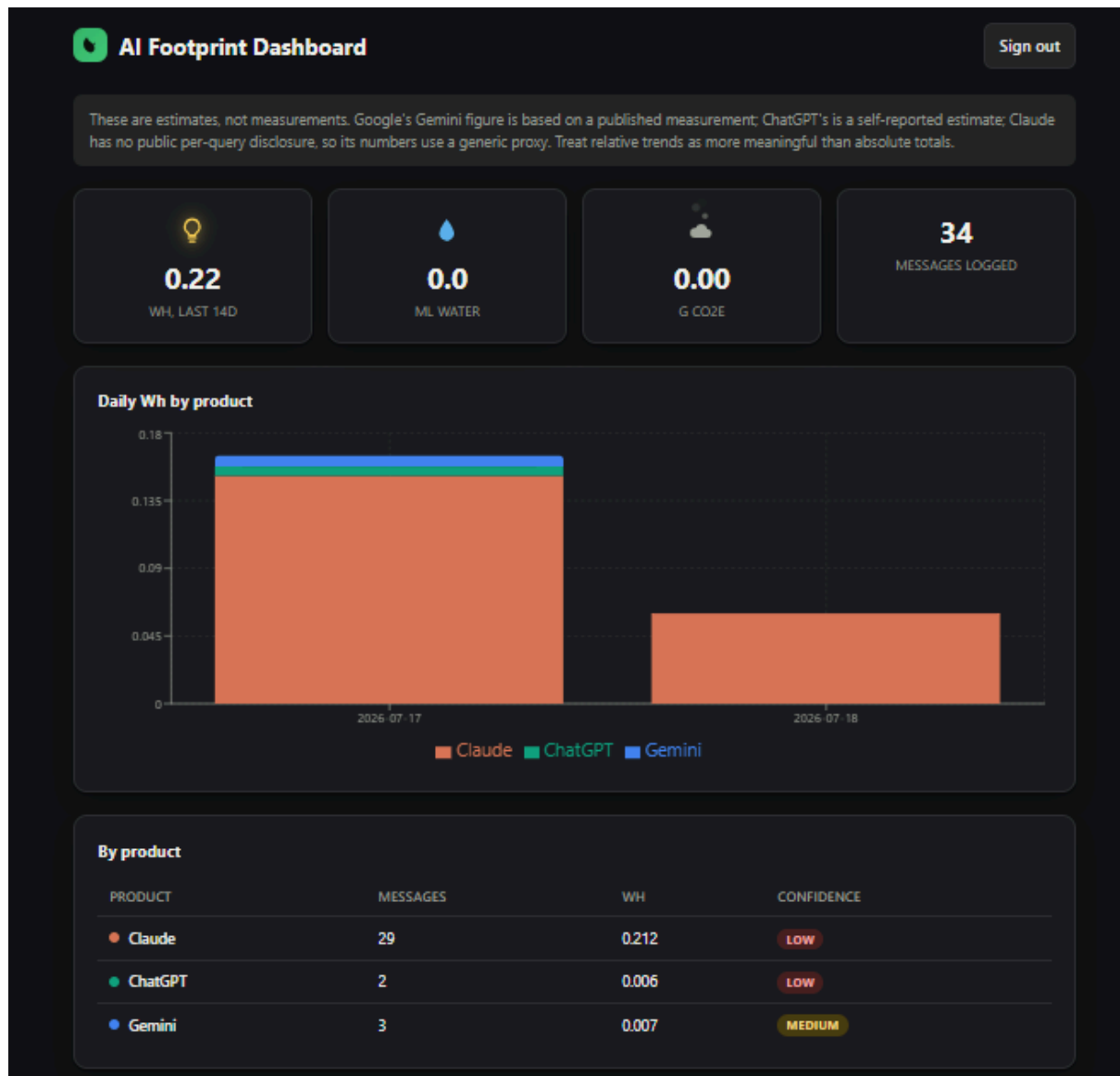
The system as a whole is really two loosely connected pieces, addressing the demand and supply sides of AI's footprint respectively. The inference tracking extension runs client-side in the browser. The edge computing platform operates at the model-deployment layer. They share the same estimation logic underneath: the same published coefficients for energy, water, and carbon intensity per unit of compute feed both the extension's real-time estimates and the platform's case-study projections. Even so, they are built to work independently of each other, so the extension delivers value now while the edge platform gets developed further.

4.2 Inference Tracking Extension: Process Flow

Figure 1. *Interaction Detected* → *Metric Estimation* → *Coefficient Mapping (energy / water / carbon)* → *Real-Time Display* → *Historical Logging* → *Equivalence Framing*

When someone interacts with a supported AI platform, the extension picks up on the interaction and estimates its computational scale from length and complexity. That estimate gets mapped to energy, water, and carbon figures using the coefficients from Chapter 2, shown to the user right away, logged locally to build up a daily, weekly, and all-time history, and finally reframed as something relatable, a fraction of a phone charge, millilitres of water, grams of CO₂, so the number means something without requiring anyone to interpret raw units.





4.3 Edge Computing Platform: Proposed Architecture

Figure 2. *Data Upload → Data Preprocessing → Model Selection → Model Management → Edge Optimisation → [Feed to on-device chips, e.g. NVIDIA Jetson-class hardware] OR [Download package for offline edge upload via USB]*

The platform's core job is optimising trained models for edge deployment. A developer uploads data, it gets preprocessed and used to select and manage an appropriate model, and the platform applies edge optimisation (quantization, pruning, or conversion to a lightweight runtime) before the resulting model either goes straight to networked edge chips or gets packaged for download and manual, offline installation onto a target device. That second path is there on purpose, so deployment does not require a live internet connection, which matters given the accessibility goals set out in Chapter 1.

4.4 Practical Example: Crop-Health Monitoring on a Raspberry Pi

The following worked example illustrates the platform's intended use, taking image-based crop-health monitoring as a representative edge-AI application relevant to energy-undersupplied, agriculturally dependent regions.

Step 1: Data Collection

A camera connected to a Raspberry Pi captures images of crops. Images are uploaded to the platform via CLI or, where the Pi is connected, directly through the platform interface.

Step 2: Data Preprocessing

Images are resized, normalised, and filtered to enhance relevant features, and labelled as "healthy" or "unhealthy".

Step 3: Model Training

A convolutional neural network is selected and trained on the labelled images within the platform's studio, which configures the model for edge efficiency from the outset.

Step 4: Model Optimisation

The trained model is converted to a lightweight format (e.g. TensorFlow Lite) sized to fit the memory and processing constraints of the target device.

Step 5: Model Deployment

The optimised model is downloaded and transferred to the Raspberry Pi via SSH, USB, or direct download, then loaded using the platform's SDK or generated code.

Step 6: Real-Time Monitoring and Updates

The Pi runs inference locally against the live camera feed, classifying crop health without an internet connection, and can be updated by feeding new data back through the platform and redeploying.

None of this data needs to leave the field to get processed, which is the same reason it is both energy-efficient (Chapter 5) and usable somewhere without reliable connectivity.

4.5 Comparative Case Study: Cloud vs. Edge Processing (Smart Agriculture Scenario)

To get a sense of the platform's potential impact before building it in full, I modelled cloud and edge processing for a representative smart-agriculture deployment: a soil-moisture sensor network used to optimise irrigation.

Assumption	Value
Data collected	1 reading per sensor per minute, ≈ 1 KB per reading
Number of sensors	100, deployed across a farm
Cloud data-centre power	400 W per server; cooling/operational overhead adds a further 400 W (2 $\times$)
Edge device power	5 W per device
Cloud data transmission	0.1 kWh per GB transferred
Edge data transmission	≈ 0.01 kWh/year for periodic summary updates only

Table 2: Smart agriculture scenario assumptions.

Cloud Processing

Component	Calculation	Result
Data transmission	$100 \text{ sensors} \times 1 \text{ KB/min} \times 60 \times 24 \times 365 \approx 52.56 \text{ GB/yr}$ $\text{GB/yr} \times 0.1 \text{ kWh/GB}$	5.256 kWh/yr
Data-centre operation	$1 \text{ server} \times 400 \text{ W} \times 2 \text{ (cooling)} \times 24 \text{ h} \times 365 \text{ d}$	7,008 kWh/yr
Total energy	$5.256 + 7,008$	7,013.256 kWh/yr
Carbon footprint (0.5 kg CO ₂ /kWh)	$7,013.256 \times 0.5$	3,506.628 kg CO ₂ /yr

Table 3: Cloud processing, annual energy and carbon breakdown.

Edge Processing

Component	Calculation	Result
Data transmission	Periodic summaries only	0.01 kWh/yr
Device operation	$100 \text{ devices} \times 5 \text{ W} \times 24 \text{ h} \times 365 \text{ d}$	438 kWh/yr
Total energy	$0.01 + 438$	438.01 kWh/yr

Component	Calculation	Result
Carbon footprint (0.5 kg CO2/kWh)	438.01×0.5	219.005 kg CO2/yr

Table 4: Edge processing, annual energy and carbon breakdown.

Factor	Cloud Processing	Edge Processing
Data transmission	High (5.256 kWh)	Minimal (0.01 kWh)
Computational power	High (7,008 kWh)	Lower (438 kWh)
Cooling systems	Significant	None needed
Total annual energy	7,013.256 kWh	438.01 kWh
Carbon footprint	3,506.628 kg CO2/yr	219.005 kg CO2/yr
Accessibility in underserved areas	Limited by network infrastructure	Highly accessible

Table 5: Summary of differences between cloud and edge processing.

Moving from cloud to edge processing in this scenario cuts projected annual energy consumption and carbon footprint by around **93.75%**, mostly from removing always-on cooling and server overhead and cutting down what actually needs to be transmitted off-site. That is in line with, and a little better than, independently published estimates that edge platforms can achieve over 90% energy savings and more than 80% reductions in carbon and water use relative to cloud generative-AI inference ^[12], and with task-offloading research reporting carbon-footprint reductions of up to 83.3% ^[13]. Having independently derived figures land in roughly the same place is reassuring, given that the platform itself has not been deployed yet.

4.6 Current Considerations for the Edge Platform

- The platform, as currently scoped, works with image data only.
- Pretrained models are currently limited to agricultural use cases (e.g. crop-health classification), pending broader dataset support.
- Edge hardware supply within target regions (affordable Raspberry Pi-class devices and accessories, for example) is a practical dependency that sits outside the platform's own scope, and would need to be addressed alongside it, particularly in the African markets

identified in Chapter 2 as having the lowest current edge-server capacity share despite the fastest projected growth.

Chapter 5: Evaluation and Results

5.1 Evaluation Approach

The two systems are at different stages of maturity, one deployed, one still a design, so I have evaluated them differently rather than forcing a single shared metric. The extension is judged qualitatively against its functional and non-functional requirements from Chapter 3, and positioned against comparable tools now on the market. The edge platform is evaluated through the case study in Chapter 4, since there is no live deployment yet to measure against. That evaluation is explicitly projected, model-based evidence rather than measured results, and a future version of this report should replace it with real metering data from an actual pilot.

5.2 Case Study Results

The Chapter 4 case study is the main quantitative result here. Its central finding, a 93.75% reduction in annual energy consumption and carbon footprint when representative smart-agriculture sensor processing moves from cloud to edge, lines up with independently published research on cloud-versus-edge environmental performance for AI inference (Chapter 2). That agreement makes a stronger case for the edge platform's viability ahead of a build, though it is worth repeating that these are scenario-specific projections, not measurements of the platform itself running in the real world.

5.3 Feature Comparison: Extension vs. Existing Market Tools (2026)

Capability	This Extension	Typical Market Tools (2026)
Real-time per-interaction estimate	Yes	Yes, in several tools
Water-footprint estimate alongside energy/carbon	Yes	Partial (most report energy/carbon only)
Local-only processing, no data collection	Yes	Varies by tool
Everyday-equivalent framing (phone charge, water bottle, etc.)	Yes	Yes, common in newer entrants
Links tracked footprint to a production-side reduction path (edge platform)	Yes	Not observed in reviewed tools

Table 7: Comparison of the inference tracking extension against comparable market tools (2026). Based on the tools reviewed in Section 2.4.

On core tracking functionality, the extension holds up fine against what is already on the market. What sets it apart is pairing tracked consumption data with an actual reduction strategy on the supply side, the edge platform, rather than stopping at measurement or offsetting.

5.4 Limitations and Current Considerations

- Footprint figures produced by the extension are estimates derived from published coefficients and interaction characteristics, not direct metering of the underlying AI provider's infrastructure; accuracy is bounded by how transparent the platforms it tracks actually are.
- No formal user study has been conducted yet on whether real-time footprint visibility measurably changes AI usage behaviour. That is priority future work, covered in Chapter 6.
- The edge platform has not been deployed beyond the modelled case study in Chapter 4; its evaluation here is projective, not empirical.
- The edge platform's current scope (image data, agricultural use cases) is narrower than its eventual target, as noted in Section 4.6.

Chapter 6: Conclusions and Recommendations

AI's energy, water, and carbon footprint has gone from a background concern to a real, worsening constraint: 448 TWh of electricity and 264 billion gallons of water in 2025 alone attributable to data centres, with AI an increasingly large driver of both ^[1] ^[2]. It is a constraint that lands hardest on regions least equipped to absorb it, where connectivity and grid power are already scarce. My conclusion from all of this is that dealing with this footprint properly means acting on both sides of it at once: making it visible to the people generating it, and moving the heaviest computation away from centralised cloud infrastructure toward the edge. The inference tracking extension delivers on the first part today, as a working, deployed tool. The edge computing platform addresses the second. It is still at the design and case-study stage, but its projected impact, a 93.75% reduction in energy and carbon for a representative scenario, holds up against both my own modelling and independently published research reaching similar numbers.

Recommendations

- Extend the extension's estimation accuracy toward token-level granularity, and broaden platform coverage beyond its current supported sites, following the approach taken by newer entrants such as token-based telemetry tools.
- Build a minimum viable version of the edge optimisation pipeline described in Chapter 4, starting with the quantization/TensorFlow Lite conversion path already scoped for Raspberry Pi-class deployment, and validate it against real, metered hardware rather than modelled figures alone.
- Conduct a user study on whether real-time footprint visibility actually changes AI usage behaviour, to test the consumption-side half of the hypothesis directly.
- Pursue partnerships with the growing African edge-computing and green-compute ecosystem, including regional infrastructure initiatives such as IXAfrica and the Africa Data Centres Association, to align the platform's rollout with existing investment rather than duplicating infrastructure.
- Explore closing the loop between the two systems: use aggregate, anonymised behavioural data from the extension to work out which edge-optimisation targets (model types, use cases) would deliver the greatest real-world reduction if prioritised first.

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